

Sathira Silva

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INTERESTS

3D Perception

Vision-Language Models

Self-supervised Learning

EDUCATION

University of Peradeniya, Sri Lanka

Nov. 2018 – Dec. 2023

B.Sc. (Hons) in Computer Engineering

GPA: 3.60 / 4.00

[\[Courses\]](#), [\[Thesis\]](#)

Class: 2nd Upper

De Mazenod College

Jan. 2003 – August 2016

G.C.E. Advanced Level

Maths - A, Physics - A, Chemistry - B

EXPERIENCE

Research Engineer

May 2025 – Current

Computer Vision Department — Principal Investigator: [Dr. Muhammad Haris Khan](#) 🔗

MBZUAI 🔗

- Lead the developmentally inspired VLM training research: BabyMind where we tried to outperform existing work CVCL via infant-inspired inductive biases.
- Contributing to a domain generalizable visual recognition research direction using baby videos.
- Contributing to developing a unified medical reasoning LMM capable of solving multiple medical-specific tasks.

Research Assistant

June 2024 – April 2025

Computer Vision Department — Principal Investigator: [Dr. Muhammad Haris Khan](#) 🔗

MBZUAI 🔗

- Contributed to developing a SOTA solution ([DPA](#)) to unsupervised fine-tuning of VLMs (CLIP) using *novel dual prototypes*.
- Fabricated a *novel multi-crop-based self-training pipeline* for fine-grained unsupervised adaptation of CLIP using fine-grained interactions between the two modalities, outperforming the SOTA.
- Worked on a *novel method for uncovering CLIP's hidden fine-grained knowledge* for unsupervised adaptation of CLIP on fine-grained datasets.

Computer Vision Research Engineering Intern

Dec. 2022 – Mar. 2023

Autonomous Vehicle R&D Division

Vega Innovations 🔗

- Contributed to the integration of a transformer architecture called [NEAT](#) into an autonomous vehicle system, by reviewing the paper and understanding its internals.
- Developed real-time computer vision solutions for autonomous vehicles on high performance GPU inference embedded systems (Nvidia DRIVE PX2 / Jetson TX2).

Teaching Assistant: Programming Methodology (CO222) 🔗

May 2021 – Sep 2021

Department of Computer Engineering

University of Peradeniya

- Supervised weekly 2hr long online lab sessions.
- Created questions for online quizzes based on the C programming Language.
- One-on-one sessions with students to tutor them on the C programming language concepts.

SELECTED ACHIEVEMENTS

ACES Coders v7.0 | An inter-university algorithmic programming competition organized by of UoP

2020

Rank - 14 / 100+

Team Name: *bitLasagna*

IEEEExtreme 14.0 | 24-hour global algorithmic programming competition

2020

Country Rank - 2, Global Rank - 68 / 7000+ 🔗

Team Name: *InterGreat*

ICDS Mini Hackathon An inter-university Data Science Hackathon	2021
Rank - 5 / 100+ 	
Team Name: <i>bitLasagna</i>	
IEEEExtreme 16.0 24-hour global algorithmic programming competition	2022
Country Rank - 27, Global Rank - 427 / 6373 	
Team Name: <i>bitLasagna</i>	
ACES Coders v9.0 An inter-university algorithmic programming competition organized by of UoP	2022
Rank - 2 / 100+ 	
Team Name: <i>bitLasagna</i>	

ONGOING PROJECTS

Monotone-Order Strctural Sketch (MOSS) for Domain-Generalizable Chest X-ray Di-agnosis	Current
Index Terms: Domain Generalization, Medical Imaging, Image Harmonization	<i>Research</i>
<i>Technologies: PyTorch, Distributed Training</i>	
Developing a plug-and-play chest X-ray harmonization framework that develops an efficient structural encoding with semantic decoder adapters to preserve pathology-relevant evidence while reducing scanner- and site-specific style variation.	
MedCache: Caching for Efficient Medical World Model Inference	Current
Index Terms: Medical World Models, Video Generation, Diffusion Caching	<i>Research</i>
<i>Technologies: PyTorch, Distributed Inference, VBench / Med-VBench</i>	
Developing a caching-based inference framework for medical world models to reduce video generation latency while preserving clinical visual fidelity and temporal consistency.	

PUBLICATIONS

S2TPVFormer: A Spatiotemporal Approach to Tri-Perspective Representation for 3D Semantic Occupancy Prediction   	Dec. 2024
In AAAI 2025 Machine Learning for Autonomous Driving (ML4AD) Workshop – <i>Best paper award</i>	
<i>Authors: Sathira Silva*, Savindu Wannigama*, Gihan Jayatilake, Muhammad Haris Khan, Roshan Ragel</i>	
microCLIP: Unsupervised CLIP Adaptation via Coarse-Fine Token Fusion for Fine-Grained Image Classification  	Dec. 2025
Accepted at ACL (Findings) 2026	
<i>Authors: Sathira Silva, Eman Ali, Chetan Arora, Muhammad Haris Khan</i>	
Objects Before Words: Object-First Inductive Biases for Grounding Language in Child-View Video	Feb. 2026
Accepted at CogSci 2026	
<i>Authors: Sathira Silva, A. K. Gebreselasie, M. Umer Sheikh, Kartik Kuckreja, Daniel Harari, M. H. Khan</i>	
Towards Fine-Grained Adaptation of CLIP via a Self-Trained Alignment Score  	July 2025
[Oral] In 2026 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)	
<i>Authors: Eman Ali, Sathira Silva, Chetan Arora, Muhammad Haris Khan</i>	
DPA: Dual Prototypes Alignment for Unsupervised Adaptation of Vision-Language Models   	Oct. 2024
In 2024 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)	
<i>Authors: Eman Ali, Sathira Silva, Muhammad Haris Khan</i>	
Learning Visual Referents from Noisy Child-view Videos	Apr. 2026
Under Review in ACM Multimedia 2026	
<i>Authors: Abrham Kahsay Gebreselasie, Sathira Silva, Michael Sidorov, Daniel Harari, Muhammad H. Khan</i>	

TECHNICAL SKILLS

Languages: C/C++, Python, Java, HTML/CSS, JavaScript, SQL

Developer Tools: Visual Studio, VS Code, Eclipse, Jupyter Notebook, Android Studio

Technologies/Frameworks: OpennMLab, PyTorch, TensorFlow, Bash Scripting, GitHub, OpenCV, TensorFlow, ReactJS, NodeJS, Jekyll

CERTIFICATIONS

Natural Language Processing (hons) 
HSE University

Jan. 2022
Coursera

Algorithms on Graphs 
University of California San Diego

July 2020
Coursera

Data Structures 
University of California San Diego

June 2020
Coursera

Convolutional Neural Networks 
DeepLearning.AI

Feb. 2020
Coursera

Neural Networks and Deep Learning 
DeepLearning.AI

Jan. 2020
Coursera

RELEVANT COURSEWORK

Data Structures & Algorithms

Operating Systems

Software Methodology

Computer Architecture

Image Processing

Programming Methodology

Artificial Intelligence

Discrete Mathematics

Networking and Web Application Design

Probability and Statistics

REFERENCES

Dr. Muhammad Haris Khan 

Assistant Professor,
Department of Computer Vision,
Mohammed bin Zayed University of AI,
United Arab Emirates

Prof. Roshan G. Ragel 

Head of Department,
Department of Computer Engineering,
Faculty of Engineering,
University of Peradeniya, Sri Lanka